

Customer Lifetime Value & Churn Segmentation Analysis

Key Findings

- Segmented a 93,000-customer e-commerce base using RFM (Recency, Frequency, Monetary) scoring into eight behavioral segments, supplemented by a churn classification model, to identify which customers drive revenue and which are at risk of churn.
- At Risk (High Value)** customers — those who previously purchased frequently and at high value but have gone inactive — represent the single highest-priority segment: 15.55% of the customer base, yet 26.68% of total revenue (₹2.7M+).
- Average value per customer in this at-risk segment (₹188) nearly matches the business's most active customers (Champions, ₹198) — this group is not just large in scale, it is individually as valuable as the top tier, making it the clearest target for retention investment.
- Overall churn, defined as no purchase in the trailing 90 days, stands at 90.14%. This figure carries an important caveat (see Limitations): because the dataset represents a fixed historical window rather than an ongoing live business, this number reflects "percentage of customers inactive within the observed period," not a real-time churn signal comparable to a live monitoring dashboard.

Business Objective

The purpose of this analysis was to answer three specific business questions:

- Which customers contribute disproportionately to revenue, and can they be reliably identified?
- Which customers are showing signs of disengagement, and how much revenue is at risk as a result?
- What distinguishes different types of high-value customers (frequent small purchases vs. occasional large ones), and does that distinction change what retention strategy would work?

Data Source

The analysis uses the Olist Brazilian E-Commerce Public Dataset, covering orders placed between 2016 and 2018. Core tables used:

- Orders** — order-level status and timestamps
- Customers** — customer identifiers (note: `customer_unique_id` was used throughout rather than the per-order `customer_id`, which is reassigned on every order and would otherwise cause every customer to appear as a one-time buyer)
- Order Payments** — transaction value per order
- Order Items / Products** — product-category detail

Only orders with a status of "delivered" were included in customer value calculations, to avoid inflating recency or spend with cancelled or unfulfilled orders.

Methodology

1. Data Preparation (SQL) Order, customer, and payment tables were joined and filtered to delivered orders only, forming the base transaction table used for all downstream analysis. Views were materialized into physical tables and indexed on join keys to support acceptable query performance at scale.

2. RFM Scoring Each customer was scored on three dimensions, using data-set-relative quintiles (1–5 scale):

- **Recency** — days since last purchase, measured against the dataset's own most recent order date (not the current calendar date, since the dataset itself is historical)
- **Frequency** — total number of distinct orders placed
- **Monetary** — total amount spent

3. Segmentation Customers were grouped into eight behavioral segments (Champions, Loyal Customers, At Risk – High Value, At Risk – Low Value, Big Spenders (New), New/Promising, Potential Loyalist, Lost) based on combinations of R, F, and M scores. Segment logic was validated by checking that revenue share consistently exceeded customer-count share for high-value segments, and confirming no single segment absorbed an implausibly large share of the customer base.

4. Churn Classification Churn was defined as no purchase within 90 days of the customer's last order relative to the dataset's observed end date. A Random Forest classifier was trained using frequency and monetary value as features. Recency was deliberately excluded from the model's feature set, since it was used to define the churn label itself — including it would have caused the model to simply reproduce the labeling threshold rather than learn a genuine predictive signal (data leakage).

5. Validation Segment-level average revenue per customer and average order value were both calculated as revenue-weighted ratios (total segment revenue ÷ total segment orders), not as simple row-level averages, to avoid distortion from customers with very different order counts.

Limitations

- **Historical, fixed-window dataset:** The data spans a fixed 2016–2018 period rather than an ongoing operational feed. Customers who made their only purchase near the end of the observed window had little to no opportunity to make a second purchase within the window, which inflates the overall churn rate relative to what a live business would report. Churn and retention figures should be interpreted as descriptive of this dataset's specific timeframe, not as a general benchmark.
- **Low baseline repeat-purchase rate:** The dataset shows a 3.00% repeat purchase rate overall, consistent with known characteristics of the Olist dataset rather than a finding specific to this analysis.

- **90-day churn threshold is a business assumption:** This threshold was chosen as a reasonable default and could be adjusted depending on actual product purchase cycles if applied to a real, live business.